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AI Development Process

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Version History

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Reviewers

Name	Role	Date
Jason Doe	Head of IT	01/07/20XX

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Introduction

Scope

- This process covers the development activities undertaken for systems and models only. It does not cover edge device software, such as for mobile devices and tablets. Development for edge devices is covered in <<reference>> Edge Device Development Process.

Background

- The concept of secure development, whereby techniques are adopted to build security into systems, rather than bolt on after the fact, has been in place for many years, promoted by industry standards such as ISO 27001. With the emergence of AI systems, it is necessary to also consider other aspects, most notably:
    - **Ethical programming**: satisfying AI perspectives such as bias and explainability.
    - **Efficient programming**: taking into consideration the impact of the inevitably intensive nature of AI computation on climate change by developing more efficient code.
- System development within <<organization>> is centered around the use of DevOps within the Azure environment, utilizing <<version control system>> and <<ticketing system>> for change management (see also the IAMS0007 Change Management Process).
- This document sets out the development approach adopted within <<organization>>, covering:
    - Agile development and the use of DevOps
    - Governance and segregation, covering segregation of environments through the use of DevOps, version control, security of code, knowledge management and vulnerability management.
    - Quality management and quality gates
    - Data quality (model test and validation)
    - AI development process steps:
        - Request / requirement gathering, feasibility and prioritization (when will the change or system be implemented)
        - System analysis and risk review
        - Design
        - Coding
        - Unit/system testing

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        - Integration and integration testing
        - Acceptance testing
        - Release process
        - Post-release verification
    - AI development process monitoring
    - Internal audit support.
- The path through the process depends on the type of software being developed / changed:
    - **Models**: core AI model software which is used by systems to deliver a service to a customer, such as the predictive analysis engine. Note that there can be multiple instances of any single model, trained differently to meet a particular system's needs.
    - **Systems**: AI products provided to customers or used internally, such as Healthcare AI, or non-AI systems, such as the internally developed <<organization>> resource management system.

Agile Development and DevOps

- <<organization>> utilizes agile development techniques using DevOps to break down large, complex development tasks into smaller work items. This delivers higher quality due to the greater testing density, that is comprehensive testing undertaken at each step on smaller segments of code.
- Through DevOps, the entire development and operation lifecycle can be managed, utilizing continuous integration, continuous delivery and continuous deployment, with development activities broken down into discrete two-week Sprints, as depicted below.

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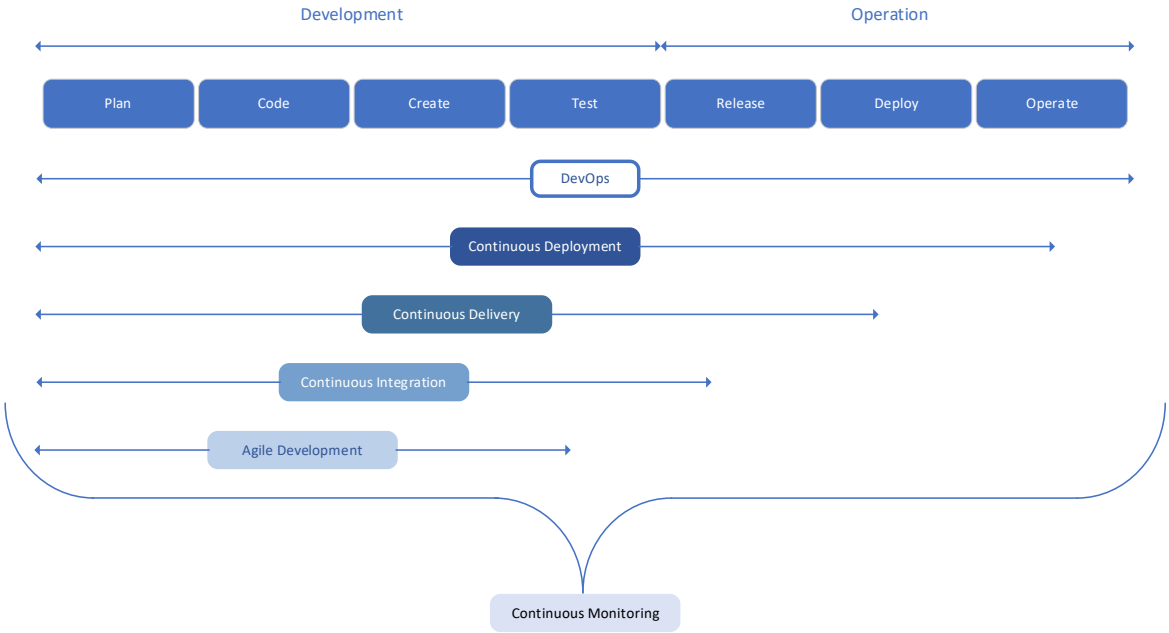

Continuous Development

- The iterative process for developing software involving continuous integration, continuous testing, continuous delivery, and continuous deployment. Within <<organization>> this includes the process steps from requirements analysis to unit testing.

Continuous Integration

- Continuous integration (CI) extends continuous development to include system integration, whereby code is regularly merged into a shared repository where those updates are tested. CI ensures the most up-to-date and tested code is always readily available to developers. Within <<organization>> this includes the process steps from requirements analysis to integration testing.

Continuous Testing

- Continuous testing extends CI to encompass all aspects of testing prior to release into production. Testing can surface bugs and particular aspects of the product that may need fixing or improvement, and can be pushed back to the development stages for continued improvement. Within <<organization>> this includes the process steps from requirements analysis to validation.

Continuous Delivery

- Continuous delivery extends continuous testing to encompass packaging code for delivery to enable rapid release cycles. The release process is largely automated (through merging branches into the main branch, so that new builds can be released at the click of a button. Within <<organization>> this includes the process steps from requirements analysis to release.

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Continuous Deployment

- Continuous deployment allows for the fully automated release of code (the fully automated version of continuous delivery) with no human (i.e., manual) intervention. Given the need for human oversight in the development of AI systems (and models), this has not been adopted within <<organization>>. This reflects the corporate culture of <<organization>> and the need to demonstrate human oversight to customers.

Continuous Monitoring

- Continuous monitoring allows for the recording of metrics throughout the development pipeline. This allows IT operations to identify issues and notify developers in real time, and for the continuous improvement of the AI development process.

Governance and Segregation

- An agile development approach through DevOps does not consist of development, test and production environments as would be present in a traditional development process. Instead, source code is held within DevOps, within which all development and almost all test activities are undertaken, and from within which production builds are produced for release into production environments. The exception being acceptance testing, which is performed in an acceptance test environment.
- Production environments may be within <<organization>>, software as a service (SaaS) solutions on Azure virtual environments (either shared by or dedicated to individual customers) or on customer-operated systems.
- Details of the DevOps architecture and the various environments in operation are set out in the <<reference>> <<organization>> DevOps architecture document.

Code Security and Version Control

- Source code is held within DevOps, with access restricted to member s of the AI Development Team. Any member of the AI Development Team can check out code to undertake development work. However, only senior developers are able to build systems and release of built systems into production environments is restricted to the Head and Deputy Head of AI Development only.
- Source code versioning is handled automatically by DevOps, which maintains an audit trail of all modifications and the ability to back-out changes to individual lines of code. Every step of the life of a (<<ticketing system>>) ticket and the Story is tracked using the “Status” field from the moment it is created until it is closed. This will also include whether or not the ticket was closed, re-opened and re-closed.
- Source code is held in a source code branch within DevOps. The source code branch for a given release of software is set up by the DevOps manager.

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- Tools employed by the AI Development Team undergo an approval process before they are used by members of the AI Development Team, as set out in the <<reference>> AI Tool Approval Procedure. Members of the AI Development Team use tools approved through this procedure only.

Competency and Knowledge Management

- Competency of Development Team members is tracked through the AIMS0020 Competency Matrix, managed by HR, which is supplemented by the <<reference>> AI Development Team Skills Matrix, maintained within the AI Development Team.
- Within the skills matrix, the required skills and training are set out for each role within the AI Development Team, such as programmer, tester, senior developer, etc. Individuals are expected to acquire the necessary skills for a role, for example through training prior to progressing into that role.
- Individuals within the AI Development Team are encouraged to seek appropriate industry certifications related to their current and potential future roles. Having obtained certifications, individuals are expected to maintain CPD points to retain these certifications.

<u>Knowledgebase (wiki)</u>

- The AI Development Team maintains a knowledge base in the form of a wiki, which includes knowledge articles covering (but not limited to):
    - Detailed development procedures
    - Development tools
    - Release notes for system (and model) releases
    - Technical information
    - Design and architectural approaches.
- Each knowledge article is assigned an owner (the author) who is responsible for maintaining the article until it is retired. Retired articles are archived when no longer required or superseded by new materials.
- The Knowledgebase is managed by the Knowledgebase Manager, who ensures that authors review and update their articles on a regular basis and manages the archive of retired articles.

Vulnerability Management

- The AI Development Team participates in the remediation of vulnerabilities identified through the regular vulnerability scans and penetration tests undertaken by the Cybersecurity Team. For details of the vulnerability scanning and penetration testing currently undertaken, refer to the <<reference>> Vulnerability Management Process.
- Vulnerabilities identified by Cybersecurity related to AI systems (and models) developed by the AI Development Team (details of the current systems and models are in the <<reference>> AI Product Catalogue) are fed to the AI Development Team in vulnerability notifications (see below for the processing of these).

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- The Cybersecurity Team host a regular (currently weekly) call to discuss identified vulnerability and remediation progress. The Head of AI Development (or Deputy Head of AI Development) attends all calls, supported by senior developers on an as needed basis.

Quality Management

- Quality management within the AI Development Teams is twofold:
  - **Quality gates**: quality checkpoints at key points in the development lifecycle, covering information security, AI perspectives (ethics) and code efficiency.
  - **Test data quality, model training and model validation**: activities undertaken within the AI Development Team to support the Data Management Team's data quality processes, as well as model training and validation undertaken using training and validation data provided (and managed) by the Data Management Team, per the AIMS0008 Data Management Process.

Quality Gates

- Seven quality gates have been defined:
  - **Gate 1 – Requirements**: covering inclusion of information security and AI perspectives requirements during system requirements specification
  - **Gate 2 – Design**: ensuring the consideration of information security and AI perspectives as part of the design process
  - **Gate 3 – Coding**: ensuring the consideration of efficiency in the code produced, to minimize the impact on climate change
  - **Gate 4 – Unit Testing**: ensuring that all information security and AI perspectives considered in the design stage are addressed in unit testing
  - **Gate 5 – Integration**: ensuring consideration of information security, AI perspectives and efficient coding as part of the integration of modules and with live code libraries
  - **Gate 6 – Integration Testing**: ensuring that information security, AI perspective and efficient coding considerations during integration are reflected in integration testing
  - **Gate 7 – Acceptance**: ensuring the consideration of information security and AI perspectives from the requirements stage are covered during acceptance testing

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- Further details of the quality management process are set out in the <<reference>> Quality Manual. Details of each quality gate are set out in the description of the relevant AI development process step below.
- The information gathered in the quality gates are intended for use by the Quality Assurance Team, working primarily with the AI Development Team and Data Management team to identify issues with the AI Development Process (see also AI Development Process Monitoring).

Test Data Quality, Model Training and Model Validation

- Model training and validation is undertaken only for AI models, not for systems. It is important to note that systems can include AI systems, for example Healthcare AI which whilst being an AI system is little more than a user front end for a product specific instance of the predictive analysis engine, in which the AI model exists. As such, a change to the Healthcare AI system would be processed as a system change, whilst a change to the predictive analysis engine would be processed as a model change.
- Model training is undertaken for any new AI model or changes to existing AI models. A change to an AI model's code can have a profound impact on the model's behavior, so any previous training will need to be repeated. Model training is undertaken by the Data Management Team.
- Model validation, through which model performance is optimized by fine tuning the hyperparameters, is similarly undertaken for new models and instances of models, but also whenever a model is changed. Model validation is performed using validation data managed by the Data Management Team.
- Data employed for model testing throughout the AI development process is sourced from the Data Management Team, in accordance with their Data Management Process.
- Model validation information is passed to the Data Management Team as feedback on their data quality processes.

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AI Development Process

Process Flow

- The development process is illustrated below.

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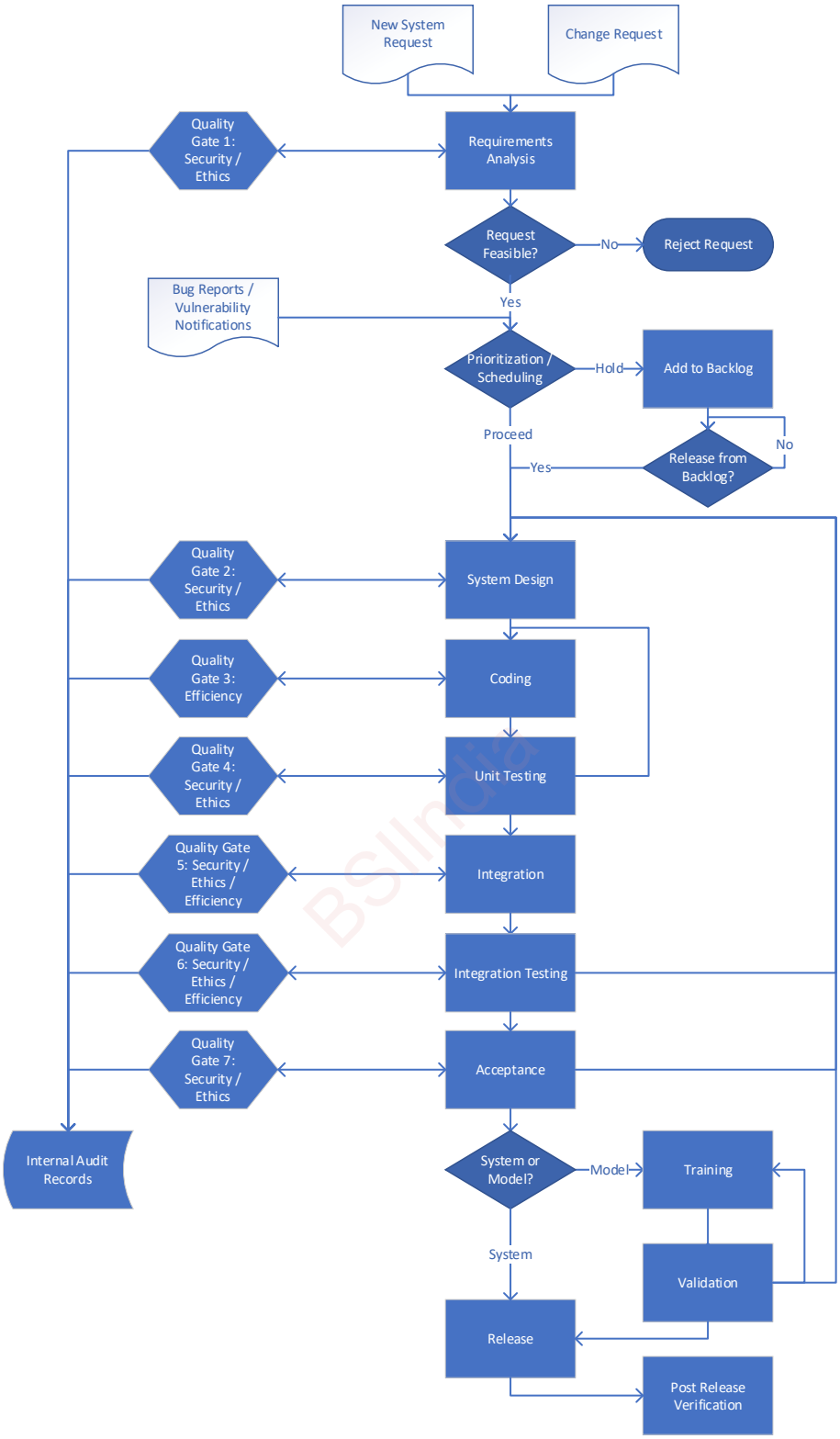

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AI Development Process Steps

- The induvial steps in the AI Development Process are as follows:

Request / Requirement Gathering

- The AI Development Team receives new system requests and change requests (for existing systems) through the <<ticketing system>> Requests will already have been approved within <<ticketing system>> in accordance with the AIMS0007 Change Management Process.
- The AI Development Team engages with various stakeholders across <<organization>>, including the change requester and senior management within their business area to draw out the detailed requirements, as follows:
    - **Change Requester / Business Management**: business requirements
    - **Cybersecurity**: information security requirements, including:
        - The expected classification of the information to be handled by the new system or as a result of the change (as defined in the <<reference>> <<organization>> Information Classification Policy)
        - Access restrictions required for the information to be handled, including the use of role-based access as necessary and expected security related roles for the system
        - Authentication mechanism required (password, multi-factor authentication)
        - Privacy requirements
        - Encryption requirements
    - **AI Ethics Team**: requirements related to the AI perspectives relevant to the change / system, from those set out in ISO 42005:
        - Accountability
        - Transparency
        - Fairness and discrimination
        - Privacy
        - Reliability
        - Safety
        - Explainability
        - Environmental impact
    - **Legal Team**: legal and regulatory requirements
    - **Risk and Compliance Team**: regulatory requirements
    - **AI Strategy Team**: alignment with <<organization>> AI strategy
    - **Data Management Team**: impact on and requirements related to data management
    - **Model Testing**: impact on and requirements related to training data
    - **Quality Assurance Team**: data quality requirements.

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- The AI Development Team is required to engage with each team set out above, who will provide relevant requirements or indicate that there are no relevant requirements from their area.
- Having engaged with other teams per above, the AI Development Team will then establish:
    - Efficiency requirements
    - Resilience requirements
    - Whether there is a need to handle malicious code, e.g., scanning of data/documents uploaded to system
    - Processing of output, including where output is to be sent, by what mechanism and any security / access control requirements associated with this by reference to the <<reference>> <<organization>> System Access Control Standard
    - Input verification and validation requirements.
- The AI Development Team holds a weekly meeting during which tickets are assigned and team members who will conduct code reviews are identified for the release in question.

<u>Quality Gate 1</u>

- Requirements, including information security and AI perspectives requirements have been produced.
- All stakeholders have been engaged.
- Requirements have been documented as part of the <<ticketing system>> ticket or in a separate requirements specification.

Feasibility and Prioritization

- Having gathered requirements regarding a new system or system change, the AI Development Team undertakes a feasibility analysis to determine whether the proposed system / change is financially, practically and technologically feasible. The analysis is undertaken in accordance with the <<reference>> Feasibility Analysis Procedure, with budget estimates for the system/change produced by senior developers. If the request is deemed not feasible then the request is rejected, and this process terminates.
- Bug reports and vulnerability notifications are mandatory so do not need to be subject to a feasibility analysis, nor is requirements gathering required, since these are fixes for identified problems with specific issues to be resolved.
- Weekly planning meetings are held for the AI Development Management Team, chaired by the Head / Deputy Head of AI Development and including all senior developers (team leaders), to consider how to structure work:
    - New system developments, or large changes resulting in system version increments are handled as single projects and are broken down into smaller

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work items. An 'Epic' is created (within DevOps) for each such project, for which the required work is broken down into one or more Stories, and each Story broken down into one or more Tasks. Each Task thus defined must be achievable within a single two-week Sprint.
    - Work to address vulnerabilities is entered into DevOps as a new Story within the pre-defined 'Vulnerability' Epic, broken down into one or more Tasks for inclusion into Sprints.
    - Work to address bug reports is entered into DevOps as a new Story within the pre-defined 'Bug fix' Epic, broken down into one or more Tasks for inclusion into Sprints.
    - Smaller changes are entered into DevOps as a new Story within the 'Changes' Epic, and are broken down into one or more Tasks for inclusion into Sprints.
- Along with the creation of an Epic, the DevOps Manager creates a new branch for the work associated with the work being undertaken and leading to a new release of the system concerned.
- As part of the planning meeting, the AI Development Management Team considers the current workload of each development team and the resources required for each new work item. Work that can be slotted into existing workloads will be scheduled for a forthcoming Sprint (a rolling schedule of the next four Sprints is maintained). Any work items that cannot be entered into the Sprint schedule are placed into the backlog for inclusion in a future Sprint.
- The exception to this is for new system developments, where functional and design specifications need to be defined before work can be broken down into Stories and Tasks (see below). Once this design work is completed the AI Development Management Team will consider allocation of work in the next weekly planning meeting.
- Work items are prioritized as follows:
    - Vulnerability reports are entered into the next Sprint for immediate resolution
    - Depending on criticality, Bug reports are entered into the Sprint schedule or added to the backlog
    - Tasks for change requests and new system developments are prioritized based on business need, cost and complexity, with individual Tasks allocated in sequence in the Sprint schedule or added to the backlog.
- As part of the weekly planning meetings, the AI Development Management Team reviews the backlog to determine whether workload permits taking Tasks from the backlog. Tasks are taken from the backlog with highest priority first and in chronological order (first in first out for a given priority level). High priority Tasks in the backlog are undertaken in preference to lower priority Tasks new that week.

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- Tasks are assigned to developers based on their skills and competencies, confirmed against the <<reference>> Skills Matrix and AIMS0020 Competency Matrix.
- Developers draw Tasks allocated to them from the current Sprint, in the sequence defined in the Sprint (it may be that multiple Tasks need to be completed in a specific order). All of the Tasks for the same release will have a tag identifying it as part of the same release (e.g., "20XX R2").
- Each developer is responsible for referencing the Story / Task and verifying that the work to be performed is understood. Any clarification must be undertaken with senior developers and by reference to the relevant AI Product Lead, as necessary before progressing with the Task.
- The AI Development Team, in collaboration with the Cybersecurity Team, AI Ethics Team and HR is responsible for identifying any impact on the annual information security and AI training undertaken by <<organisation>> staff, so that training courses can be updated as needs be.

Design

- The system design work required for a change / system will vary depending on the size of the work being undertaken, as follows:
    - **New system**: A full system design exercise is undertaken outside of the Sprint schedule. Typically, a system design exercise will take many weeks to complete, requiring several review cycles. Upon completion of the system design, it will be considered in the next weekly planning meeting, per above. The following outputs are required from the design process:
        - System function specification: one overall document for each system
        - System design specification(s): one or more documents for each system, or system component
    - **Changes, bug fixes and vulnerability notifications**: The existing system design is reviewed and updated as necessary to reflect the change required. For larger such work items, the design work may be placed as a separate Task into a Sprint, with subsequent Tasks sequenced for the completion of coding, etc.
- The developer must ensure that all requirements specified have been addressed by the system design.
- The system design needs to consider the secure and ethical system architecture and engineering principles, as set out in the <<reference>> System Architecture and Engineering Principles.
- The system design is subject to peer review by a developer unconnected with the work being performed.
- Having completed design work the developer updates the Task status and commences coding.

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- Quality Gate 2
- Functional specification and design specifications have been developed.
- Functional and design documentation has been approved by the Head / Deputy Head of AI Development and AI Product Leads.
- All requirements are addressed by the system design, as verified through peer review.
- Task status is updated.

Coding

- The implementation of the software design commences with checking out the relevant source code within DevOps into the branch for the release concerned and modifying the code as necessary, or by the writing of new code, in the relevant programming language (the language used can vary from system to system, depending on the nature of the system).
- Where multiple pieces of code need to be checked out for a given change, all of the code concerned is placed into the same branch for the work being performed. Note that code associated with multiple Tasks can be placed into the same branch.
- Coding must be undertaken in accordance with the secure and efficient coding principles as set out in the <<reference>> Secure AI Coding Principles.
- Developers are expected to alpha test their own code as they develop it, but the code is also subject to peer review by a developer unconnected with the work being performed. The code reviewed will be allocated for the particular system release during requirements gathering. The peer review must encompass the efficiency of the code produced.
- Having completed coding for a Task and peer reviews undertaken, the developer updates the Task status and make the code available for unit testing.

<u>Quality Gate 3</u>

- Results of alpha testing and peer reviews have been documented.
- Task status is updated.

Unit Testing

- Code produced for individual Tasks is subject to standalone unit testing. The unit testing must be performed by a different developer than the person who wrote the code, i.e., the tester.
- A unit test specification is produced by the tester, which sets out the test steps and the expected results for each step. The tester references the Story / Task to ensure the scope of the testing is appropriate.
- Testing must include testing of any security functions and AI perspectives. The testing of AI perspectives should be undertaken in accordance with the <<reference>> AI Testing Strategy, and can draw from the library of AI perspectives

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tests held on the AI Development SharePoint. Note that for models, AI perspectives are covered more extensively as part of validation testing.

- Testing is undertaken in the tester's own test environment.
- Test results are recorded against the test specification steps and the result summarized in the unit test report.
- If code fails unit testing, the Task is updated to failed unit testing and allocated back to the developer who wrote the code for corrective action. Unit test failure will signify a defect against the system design, so will require re-coding rather than re-design.
- The developer must record any remedial work being undertaken in the work schedule for their team, so that this can be considered as part of the assessment of workloads in weekly planning meetings during prioritization.
- Once unit testing has been completed successfully, the Task status is updated and the code is made available for system integration.

Quality Gate 4

- Unit test specification has been produced.
- Unit tests take into consideration information security and AI perspectives.
- Unit test report has been produced and testing has been completed successfully.
- Task status is updated.

Integration and Integration Testing

- Upon completion of unit testing, code is made available for system integration, whereby modified / developed code is integrated with all other code for the system and with standard code libraries to form the complete system. Note that at this time, the integrated system remains within DevOps, i.e., is not running on target hardware. Integration is undertaken as follows:
    - Where code is part of a wider piece of work, perhaps a new system development, other code required for the system may not have been completed as yet. In this case, integration is put on hold pending completion of all code required for the request/bug report/vulnerability notification.
    - Where changed code completes the work for a request/bug fix/vulnerability notification, system integration can be commenced.
- System integration pulls any associated code from the DevOps source code repository for the given software release and integrates it with any code from standard libraries to form the complete system. The integration is accomplished by merging the system's release branch with other branches containing the standard libraries or other code as necessary. System integration should include the latest versions of libraries that have been reviewed for efficient coding practices (see <<reference>> Integration Library Management Process for details of the review of libraries for efficient coding).

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- An integration test specification is produced, which sets out the test steps and the expected results for each step. The tester references the Epic / Story to ensure the scope of the testing is appropriate.
- Testing must include testing of any security functions and AI perspectives. The testing of AI perspectives should be undertaken in accordance with the <<reference>> AI Testing Strategy and can draw from the library of AI perspectives tests held on the AI Development SharePoint. Note that for models, AI perspectives are covered more extensively as part of validation testing.
- Integration testing should also include exercising code to confirm the use of efficient coding techniques, for example comparing execution times between successive generations of the same code (as a general rule, fast code is efficient code).
- Test results are recorded against the test specification steps and the result summarized in the integration test report.
- If code fails integration testing, the Task is updated to failed integration testing and allocated back to the developer who wrote the code for corrective action. Unlike unit testing, failure in integration testing is likely to indicate a fundamental system design error, so remedial work will encompass re-design as well as re-coding.
- The developer must record any remedial work being undertaken in the work schedule for their team, so that this can be considered as part of the assessment of workloads in weekly planning meetings during prioritization.
- Once integration testing has been completed successfully, the Task status is updated and the code is made available for acceptance testing.

<u>Quality Gates 5 and 6</u>

- Confirmation that latest libraries, reviewed for efficient coding have been used during integration.
- Integration test specification has been produced.
- Integration tests take into consideration information security and AI perspectives.
- Integration tests evaluate code efficiency through performance testing.
- An integration test report has been produced and testing has been completed successfully.
- Task status is updated.

Acceptance Testing

- Following successful completion of integration testing, code is made available for acceptance testing, which is managed by the Acceptance Test Manager within the AI Development Team. Acceptance test environments are established on Azure virtual servers for all instances of production systems, i.e., where a system is provided to multiple customers, an acceptance environment is in place for each customer. Where a system is to be deployed to a customer, whether internal or external, the

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Acceptance Test Manager works with the IT Team to set up a new acceptance test environment for the new instance.

- The Acceptance Test Manager builds an executable image from the code branch (containing the code requiring acceptance testing) and places the image into the relevant acceptance test environment.
- The Acceptance Test Manager works with the relevant AI Product Manager and the (internal or external) customer to identify which individuals will be undertaking the acceptance testing, and works with the IT Team to set up access to the acceptance test environment.
- The AI Product Manager and the Development Team (usually the integration tester and responsible Senior Developer) produce an acceptance test plan and acceptance test specification, which sets out the test steps and the expected results for each step. The development team references the Epic to ensure the scope of the testing is appropriate.
- Testing must include testing of any security functions and AI perspectives. The testing of AI perspectives should be undertaken in accordance with the <<reference>> AI Testing Strategy and can draw from the library of AI perspectives tests held on the AI Development SharePoint. Note that for models, AI perspectives are covered more extensively as part of validation testing.
- The acceptance test plan and acceptance test specification must be agreed with the customer prior to the commencement of acceptance testing.
- Test results are recorded against the test specification steps and the result summarized in the acceptance test report.
- Findings from acceptance tests are assigned priorities. Those of priority 1 and 2 must be addressed prior to deployment. Those items of priority 3 and above can be addressed after deployment. Note that for models, findings are always deemed priority 1.
- If priority 1 or 2 findings exist, the Task is updated to failed acceptance testing and allocated back to the developer who wrote the code for corrective action. As with integration testing, failure in integration testing is likely to indicate a fundamental system design error, so remedial work will encompass re-design as well as re-coding.
- The developer must record any remedial work being undertaken in the work schedule for their team, so that this can be considered as part of the assessment of workloads in weekly planning meetings during prioritization.
- The Acceptance Test Manager monitors remediation of the findings and will arrange re-testing once remediation is completed.
- Once all priority 1 and 2 defects have been resolved and re-testing has been completed, the Acceptance Test Manager works with the associated AI Product

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Manager to obtain customer approval. Note that for models, acceptance testing, the customer is always internal (the AI model sponsor within the business) and is limited to ensuring that functionality has been included, since further training and validation is required before the model is made operational.

- At this point the Task status is updated and the code is made available for release, or in the case of models available for training and validation, prior to release.
- Any outstanding Priority 3 and above items are added to the system's defect list and are discussed in regular service monitoring meetings between the AI Product Manager and the customer.

Quality Gate 7

- Acceptance test plan and acceptance test specification have been produced and agreed with the customer.
- Acceptance tests take into consideration information security and AI perspectives.
- An acceptance test report has been produced and testing has been completed successfully.
- Acceptance testing results have been approved by the customer.

Training and Validation

- Where requests relate to models, the model requires training and validation prior to production operation, as follows:
    - **Training**: the process to determine or improve the parameters of a machine learning model (algorithm), using training data, whereby once trained the model will be able to process production data and come to an expected decision, for example a model trained to recognize cats will always correctly identify a cat in a picture
    - **Validation**: confirmation, through the provision of objective evidence, that a trained model has learned a task correctly. Validation must be conducted with different data than that used for training, since using training data for validation would allow the model to 'remember' a specific case, rather then correctly process new production data.
- Training and validation are undertaken by the Data Management Team. Once the model has completed acceptance testing, the associated senior developer works with the IT Team and the Data Management Team to establish a 'pseudo-live' Azure virtual server, with access set up to allow access to the Data Management Team and senior developers only.
- Once model training and validation has been completed, the Data Management Team informs the AI Development Team that the model is available for release.

Release Process

- The method of releasing code into production varies depending on whether a model or system is being released.

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Model

- Given the Data Management Team undertakes model training and validation in a pseudo-live environment, the release process for the model is to make the pseudo-live environment operational.
- The Senior Developer works with the IT Team and the relevant AI Product lead(s) to make the Azure server available for use with production systems, such as Healthcare AI. This will involve redirecting the production system(s) to 'point' to the new model in accordance with the <<reference>> Model Release Procedure.

System

- Following system testing, the Senior Developer (or Head / Deputy Head of AI Development) will build the system for production release from the code Branch, as per the process for acceptance testing, but now places the resulting executable image into the target production environment:
    - **Internal systems**: an Azure virtual server for internal use. The Senior Developer will work with the IT Team to set up the server. Access is then granted to internal users through the <<reference>> Access Provisioning Process, operated by the Service Desk.
    - **Customer SaaS systems**: a public facing Azure virtual server. The Senior Developer will work with the IT Team and relevant AI Product Lead to set up and grant access to the system. Access for each customer organization's users is set up through the <<reference>> Customer Access Management Process, operated by the Customer Service Desk.
    - **Customer operated systems**: a server operated by a customer organization. The Senior Developer works with the customer organization through the relevant customer service manager, and with the associated AI product lead to release the system to the customer and assist the customer with setting up the system. The executable build is packaged such that the system can be installed with little or no intervention. The package can be delivered to the customer through the <<organization>> secure file transferprotocol (SFTP) portal or on DVD.
- Note that internal and system and model releases are conducted through the AIMS0007 Change Management process.
- Once deployed, the DevOps Task is updated.

Post Release Verification

- Following release to production, all associated tickets, Tasks and Stories are closed by the Senior Developer (or designated deputy).
- The code Branch for the released system (or model) is merged back into the main Branch ready for subsequent releases.

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- Production is monitored for a 30-day warranty period to allow rapid resolution of issues that may materialize during production operation, after which the release is monitored as part of business as usual (BAU).
- The AI Development Team holds a post-mortem evaluation for significant software releases to discuss effectiveness of the deployment and identify areas for improvement. Process changes identified as part of this evaluation will be reflected in this document, or in other relevant processes or procedures.

AI Development Process Monitoring

- As part of continuous improvement efforts, a number of metrics have been established to allow the measurement and improvement of the efficiency, quality, and performance of the AI Development Process. The various metrics employed are categorized and sourced as follows:
  - Productivity Metrics:
    - **Lines of Code (LOC)**: Measures the size of the codebase, but should not be used as the sole indicator of productivity. Sourced from DevOps
    - **Function Points**: Quantifies the functionality delivered by the software. Sourced from <<ticketing system>> tickets and Requirements Specifications
    - **Velocity**: Measures the amount of work completed in a Sprint. Sourced from DevOps
  - Quality Metrics
    - **Defect Density:** The number of defects or bugs per unit of code (e.g., per thousand lines of code). Sourced from DevOps
    - **Code Review Feedback Time:** The time it takes to address feedback from code reviews. Sourced from DevOps audit trail
    - **Code Churn**: The frequency of code changes, which can indicate instability. Sourced from <<ticketing system>> tickets
  - Efficiency Metrics
    - **Lead Time**: The time from the inception of a software request to its delivery. Sourced from<<ticketing system>> tickets and DevOps
    - **Cycle Time**: The time it takes to complete a single cycle or task, often used in Agile development. Sourced from DevOps
    - **Throughput**: The number of features or user Stories completed in a given time frame. Sourced from DevOps
  - Customer Satisfaction Metrics:

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        - **Net Promoter Score (NPS)**: Measures customer satisfaction and likelihood to recommend the software. Sourced from Customer Service Desk feedback requests
        - **Customer Support Response Time**: How quickly customer support responds to and resolves issues. Sourced from Customer Service Desk KPIs
    - Codebase Metrics
        - **Code Complexity**: Cyclomatic complexity and maintainability index to assess code quality. Sourced from system design documentation and DevOps
        - **Code Coverage**: Estimates the proportion of code that is covered by automated testing. Sourced from DevOps and test documentation
        - **Technical Debt**: Quantifies the amount of suboptimal code that needs to be addressed in the future. Sourced from DevOps and testing records
    - Team Metrics
        - **Team Velocity**: The team's average output over a period. Sourced from DevOps and <<ticketing system>>
        - **Team Satisfaction**: Surveys or feedback mechanisms to gauge the team's morale and engagement. Sourced from the annual employee satisfaction survey.
- Senior Developers (team leaders) compile information for these metrics on a quarterly basis, which are reviewed with the Head and Deputy Head of AI Development in the quarterly AI Development Team performance review. The objective of the review is to identify areas for improvement in the execution of the AI Development Process.
- Improvement actions are reflected in this process documentation and filtered down through the AI Development Team in team meetings and huddles.

Internal Audit Support

- Information gathered from the quality gates as set out in Quality Management are stored securely to preserve as evidence in support of internal audits. Access to the information is restricted, with only read access granted to members of the AI Development Team.
- The information is made available to the Internal Audit Team during internal audits conducted as part of the annual internal audit plan.

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